

Hugging Face API Data Extraction Methodology

1. Purpose and Scope

The objective of the Hugging Face data extraction was to collect grey literature related to artificial intelligence models, embeddings, language models, and supporting tooling hosted on the Hugging Face platform. Hugging Face was selected due to its central role in the open AI ecosystem, serving as a primary distribution hub for state-of-the-art machine learning models used in both research and production environments.

Unlike GitHub, which primarily hosts source code repositories, Hugging Face provides direct access to model artefacts, documentation, and usage metadata. As such, this data source was treated as a complementary strand of grey literature, capturing deployed and reusable AI capabilities rather than development infrastructure alone.

2. Data Source

Data were collected using the Hugging Face Hub API, which exposes metadata for publicly available models hosted on the platform. The extraction focused exclusively on public model repositories, ensuring transparency and alignment with the principles of open research.

The API responses provided structured metadata, including download numbers, repository URLs, tags, and license classifications where available.

3. Search Strategy and Keyword Selection

3.1 Conceptual Focus

The search strategy was designed to identify models relevant to:

- Natural language processing
- Representation learning and embeddings
- Large language models
- Sentiment analysis and classification

- Multilingual and speech-related AI systems

These areas were selected due to their relevance to AI capability development, downstream deployment risk, and potential security or misuse considerations.

3.2 Keyword-Based API Queries

The Hugging Face API does not support complex Boolean search logic, which is comparable to that found in academic databases. Instead, a keyword-based querying approach was employed, using targeted search terms aligned with the conceptual focus areas above.

Keywords were selected to prioritise high-impact and widely reused model categories, rather than narrow or experimental tasks. Queries were executed programmatically, and results were aggregated into a unified dataset.

4. Data Extraction Process

The extraction was implemented via a scripted API pipeline that:

- Submitted keyword-based queries to the Hugging Face Hub API
- Retrieved metadata for matching model repositories
- Normalised responses into tabular format for downstream analysis

At the conclusion of this process, the dataset consisted of 218 model records.

No enrichment, filtering, or relevance scoring was applied during this stage. The dataset, therefore, represents the raw output of the API search, subject only to the constraints of the Hugging Face API itself.

5. Methodological Considerations and Limitations

Several methodological constraints were identified:

- Hugging Face metadata quality varies significantly across repositories
- Not all models provide detailed documentation or clear usage intent

- API search functionality is less expressive than GitHub's repository search

These limitations underscore the need for manual inspection and expert judgment in later stages of analysis.

6. Summary

The Hugging Face API extraction yielded a structured but unfiltered dataset of 218 publicly hosted AI models, serving as a valuable complement to GitHub repository data. By deferring filtering and validation, the methodology preserves transparency and reproducibility while enabling informed, expert-led refinement in subsequent phases of the project.